BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

String Manipulation

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. Which CPU component performs arithmetic and logical operations?

2. Why might an HDD suit a large archive?

A) No moving parts
B) Low cost per GB
C) Unlimited lifespan
D) Always fastest access

3. "An SSD is faster because its magnetic read head rotates more quickly." Correct this. Explain why.

4. A laptop is used on a bumpy bus. "Portability and durability mean the same thing." Correct this. How should this affect the storage choice?

Today's objectives

Explain the core ideas of string manipulation using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Concatenation
Joining strings creates new text; it does not add numerical values.

Length
Counting characters helps validate usernames and input.

Slicing
A slice extracts a specified portion of a string, using the selected language's indexing rules.

Data type
Characters and strings represent text, even when they contain numeric-looking digits.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Concatenation	Joining strings creates new text; it does not add numerical values.
Length	Counting characters helps validate usernames and input.
Slicing	A slice extracts a specified portion of a string, using the selected language's indexing rules.
Data type	Characters and strings represent text, even when they contain numeric-looking digits.

I Do - One worked example

A school prints a username from firstName = 'Sam' and lastName = 'Lee'. Explain how to produce 'Sam.Lee'.

1. Identify both strings and the separator.
2. Concatenate firstName + '.' + lastName.
3. The result is the string 'Sam.Lee'; do not add the text mathematically.
4. Check that the period occurs once and no spaces were introduced.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A library creates an ID from 'BK' and '204'. What string is produced?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the OCR string operation to display NETWORK.

```
word = "network"
print(word....)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State the difference between concatenation and arithmetic addition. [2 marks]

Q3. A user enters 'AB' and '12'. Write the resulting concatenated string with no spaces. [1 mark]

Q4. Explain why checking string length can support validation of a password. [3 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Concatenating strings always adds their numeric values. ☐ T ☐ F
2. Joining strings creates new text; it does not add numerical values. ☐ T ☐ F
3. Counting characters helps validate usernames and input. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 The ALU performs arithmetic and logical operations. ☐

Q2 B - Low cost per GB ☐

Q3 It does not need a moving head to seek positions on spinning platters. ☐

Q4 Durability means resisting damage; portability means being easy to carry. An SSD has no moving read/write parts, so it handles shocks better. Size and weight also matter. ☐

Code First - Model

```
word = "network"
print(word.upper)
```

Output: NETWORK

We Do - Check the model

1. 'BK204'. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Concatenation joins string content; arithmetic addition calculates a numerical sum. ☐

Q3 AB12. ☐

Q4 Length checks whether the input meets a stated minimum/maximum number of characters; alone it does not guarantee a secure password. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: